

Regression

Simple Linear Regression

Linear Regression

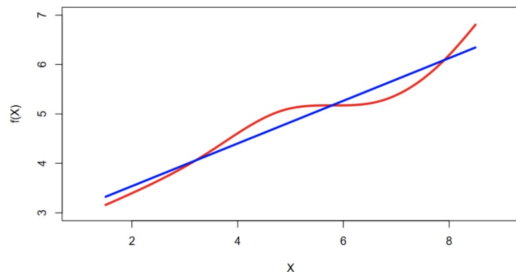
- The response variable Y is numerical;
- It assumes that the dependence of Y on the predictors $X_1, X_2, \dots, X_p$ is:
 - linear,
 - or linear after some transformation (sometimes called “*intrinsically linear*”).

Why linear regression?

- If p (the number of predictors) is large, a linear model may be complex enough;
- Interpretability;
- We can apply variable selection techniques to select features.

Simple linear regression

- Model : $Y = \beta_0 + \beta_1 X + \epsilon$;
- Data: $(x_1, y_1), \dots, (x_n, y_n)$;
- Estimated (intercept, slope) = $(\hat{\beta}_0, \hat{\beta}_1)$;
- Fitted line $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$



- We focus mainly on estimation (learning).
- We pay less attention to inference, e.g.,

$$H_0 : \beta_1 = 0 \text{ vs } \beta_1 \neq 0.$$

Least squares

- Observed values: $y_i, i = 1, \dots, n$;
- Fitted (predicted) values: $\hat{y}_i, i = 1, \dots, n$;
- Residuals: $e_i = y_i - \hat{y}_i, i = 1, \dots, n$;
- Minimize the *Residual Sum of Squares* (*RSS*):

$$RSS = \sum_{i=1}^n e_i^2;$$

- Minimizing the RSS is equivalent to

$$\min_{\hat{\beta}_0, \hat{\beta}_1} \frac{1}{2} \sum_{i=1}^n \left(y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_i) \right)^2$$

Least squares

- Let

$$g(\hat{\beta}_0, \hat{\beta}_1) := \frac{1}{2} RSS(\hat{\beta}_0, \hat{\beta}_1);$$

- To determine $(\hat{\beta}_0, \hat{\beta}_1)$, we solve the linear system

$$\begin{cases} \frac{\partial g}{\partial \hat{\beta}_0}(\hat{\beta}_0, \hat{\beta}_1) = 0 \\ \frac{\partial g}{\partial \hat{\beta}_1}(\hat{\beta}_0, \hat{\beta}_1) = 0 \end{cases}$$

which is the same with $\nabla g(\hat{\beta}_0, \hat{\beta}_1) = 0$.

- Solution $(\bar{x}, \bar{y})$ sample means):

$$\begin{aligned} \hat{\beta}_1 &= \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{(x - \bar{x})^T (y - \bar{y})}{\|x - \bar{x}\|^2} \\ \hat{\beta}_0 &= \bar{y} - \hat{\beta}_1 \bar{x}. \end{aligned}$$

R-squared

- Total Sum of Squares: $TSS = \sum_{i=1}^n (y_i - \bar{y})^2$
- Explained Sum of Squares: $ESS = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2$
- Residual Sum of Squares: $RSS = \sum_{i=1}^n (y_i - \hat{y}_i)^2$
- ANOVA identity

$$\begin{aligned} TSS &= ESS + RSS \\ (\text{Data} &= \text{Model} + \text{Error}). \end{aligned}$$

- **R-squared**: it measures the percentage of variation in Y that can be explained by the model

$$R^2 = \frac{TSS - RSS}{TSS} = 1 - \frac{RSS}{TSS}.$$

Coefficient Accuracy: Simple linear regression

Model: $Y = \beta_0 + \beta_1 X + \epsilon$

- **Standard errors** for the estimated coefficients are given by

$$SE(\hat{\beta}_1)^2 = \frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2}, SE(\hat{\beta}_0)^2 = \sigma^2 \left[\frac{1}{n} + \frac{\bar{x}}{\sum_{i=1}^n (x_i - \bar{x})^2} \right],$$

where $\sigma^2 = \text{VAR}[\epsilon]$.

- **Estimating** $\text{VAR}[\epsilon]$. In general, σ^2 is not known. The value of σ can be estimated by the *residual standard error*,

$$RSE = \sqrt{\frac{RSS}{n-2}}.$$

- **Confidence intervals.** With a probability of (approximately) 0.95, the interval

$$\left[\hat{\beta}_i - 2 \cdot SE(\hat{\beta}_i), \hat{\beta}_i + 2 \cdot SE(\hat{\beta}_i) \right]$$

will contain the true value of β_i , $i = 0, 1$.

Hypothesis Testing: Simple linear regression

Is the predictor X useful in predicting the response Y ?

- **Hypothesis testing.** This corresponds to testing

$$H_0 : \beta_1 = 0 \text{ (no relationship)}$$

versus

$$H_1 : \beta_1 \neq 0 \text{ (some relationship).}$$

- To test the null hypothesis, we compute a **t -statistic**, given by

$$t = \frac{\hat{\beta}_1 - 0}{\text{SE}(\hat{\beta}_1)}$$

Assuming that $\beta_1 = 0$, this will have a t -distribution with $n - 2$ degrees of freedom. We can compute the probability of observing any value equal to $|t|$ or larger. We call this probability the **p -value**.

Hyphothesis Testing: Simple linear regression

	Coefficient	Std. Error	t-statistic	p -value
$\hat{\beta}_0$	7.0325	0.4578	15.36	< 0.0001
$\hat{\beta}_1$	0.0475	0.0027	17.67	< 0.0001

- **Y depends on X .** We declare a relationship to exist between Y and X (reject the null hyphothesis), if **the p -value is small**.
- **Typical p -values cutoffs** for rejecting the null hyphothesis are 5% (0.05) or 1% (0.01).